

American International University-Bangladesh (AIUB)

Faculty of Science and Technology

Department of Computer Science

Web Technologies

CSE 4101 / [CSC3215]

PROJECT REPORT

Project Title:

[Online Computer Shop]

Semester: [Spring 2025–2026]

Submission Date: [May, 17, 2026]

Section: D

Submitted To:

MD. AL-AMIN

Assistant Professor, Faculty

DEPARTMENT OF COMPUTER SCIENCE

American International University-Bangladesh

Group Information

Group Number: 02

Project Title: Online Computer Shop

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Signature: _____

Declaration: We hereby declare that this project report is our original work. Any external resources or references used have been properly cited. We have not engaged in any form of academic dishonesty.

Marks Sheet

Group No.	02	Section	D
Member 1 ID	22-48356-3		

CO3: Illustrate Multi-Tier Web Application for Targeted Society					
Feature Implementation (0-3)			Feature Implementation (0-3)		Completeness (0-3)
Basic Web Security	UI (HTML/CSS)	Feature Completeness	DB	Auth (Session/cookie)	MVC
CO4: Coordinate with Group Members to Develop Multi-Tier Application					
Technical Knowledge (0-3)			Collaborative Teamwork (0-3)		Communication/Pro mptness (0-3)
JS validation	PHP Validation	Ajax/JSON	Git Contribution		VIVA
Expected Outcome (0-2)		CO3 Total	CO4 Total		Project Total

Evaluated By: _____ Date: _____

Course Teacher Signature: _____

Table of Contents

1. Group Information	2
2. Marks Sheet	3
3. Table of Contents	4
4. Chapter 1: Project Overview	5
4.1 Project Description	5
4.1 Technologies Used	5
4.3 System Architecture / MVC Diagram	6
4.4 Database Schema	6
5. Chapter 2: Git Contribution & Task Allocation	8
5.1 GitHub Repository Link	8
5.2 Git Contribution Screenshot	8
6. Chapter 3: Features, Source Code & Screenshots	10
7. Chapter 5: Conclusion & Future Work	26

Chapter 1: Project Overview

1.1 Project Description

This project is an **Online Computer Shop Web Application** where users can browse different computer components, view product details, and interact with the system based on their roles.

The system supports two main roles:

- **Admin** – manages users and system content
- **Customer** – browses products and interacts with the platform

As part of this project, I was responsible for **Task 1**, which focuses on user authentication, profile management, and the main homepage interface.

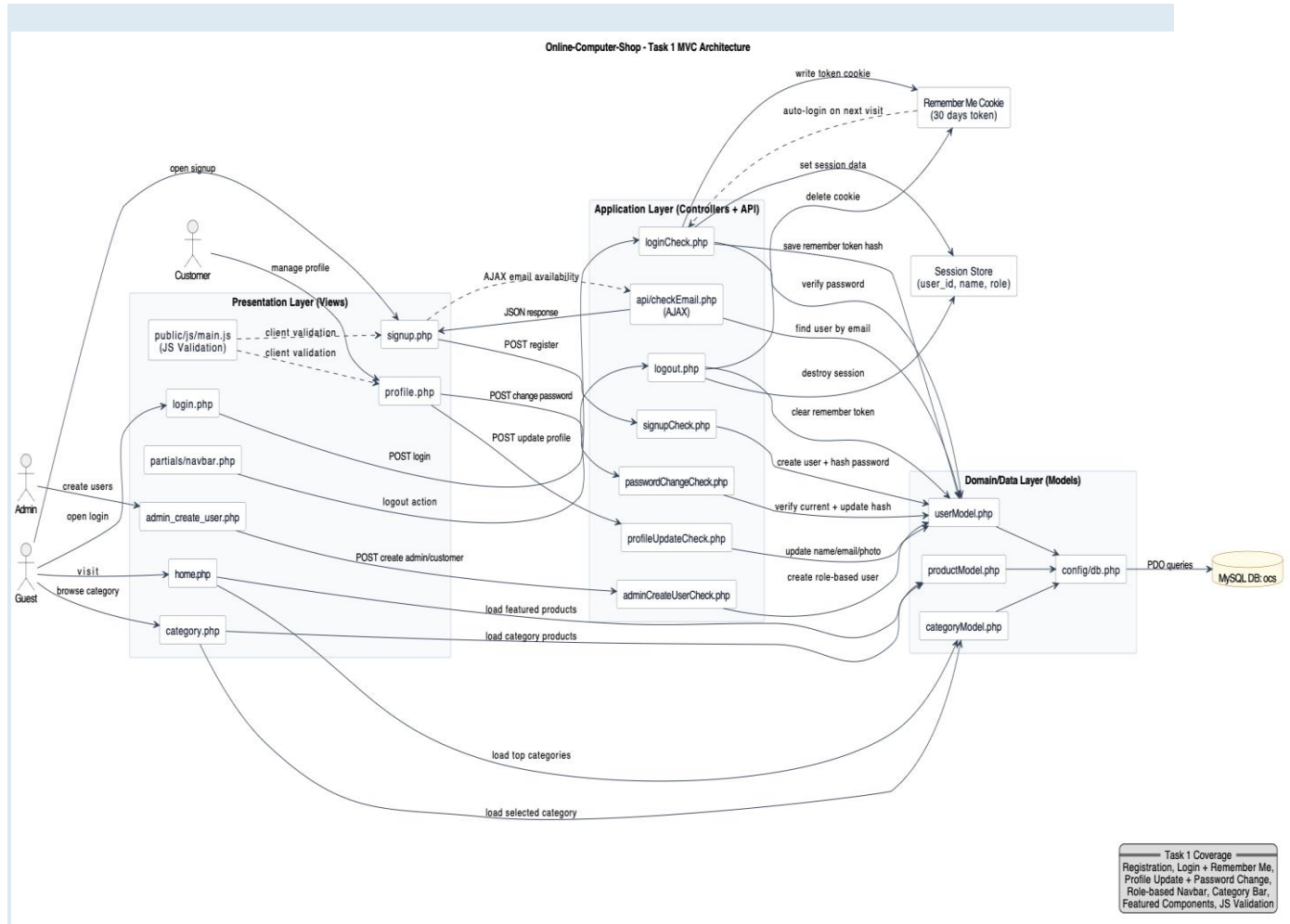
My implemented features include:

- User Registration and Login System
- Session and “Remember Me” Authentication
- Profile Management
- Dynamic Home Page
- Category Navigation Bar
- Featured Products Display

1.2 Technologies Used

Layer	Technology / Tool	Purpose
Frontend	HTML, CSS, JavaScript	UI design and validation
Backend	PHP (MVC Structure)	Business logic handling
Database	MySQL	Data storage
Auth	PHP Session & Cookies	User authentication
Version Control	Git & GitHub	Code management
Server	XAMPP / Apache	Local server

1.3 System Architecture / MVC Diagram



1.4 Database Schema

Table Name	Field Name	Data Type	Description
users	id	INT(11)	Primary Key
	name	VARCHAR(100)	User Name
	email	VARCHAR(100)	User Email
	password_hash	VARCHAR(255)	User Password
	role	ENUM('admin','customer')	User Role
	profile_picture	VARCHAR(255)	Profile Picture
	remember_token	VARCHAR(255)	Remember Token

	created_at	TIMESTAMP	Creation Time
	updated_at	TIMESTAMP	Last Update Time
categories	id	INT(11)	Primary Key
	name	VARCHAR(100)	Category Name
	parent_id	INT(11)	Parent Category ID
	created_at	TIMESTAMP	Creation Time
	updated_at	TIMESTAMP	Last Update Time
products	id	INT(11)	Primary Key
	name	VARCHAR(255)	Product Name
	description	LONGTEXT	Product Description
	manufacturer_review	TEXT	Manufacturer Review
	price	DECIMAL(10,2)	Product Price
	category_id	INT(11)	Category ID
	brand_id	INT(11)	Brand ID
	image_path	VARCHAR(255)	Product Image Path
	stock	INT(11)	Available Stock
	created_at	TIMESTAMP	Creation Time
	updated_at	TIMESTAMP	Last Update Time

Chapter 2: Git Contribution & Task Allocation

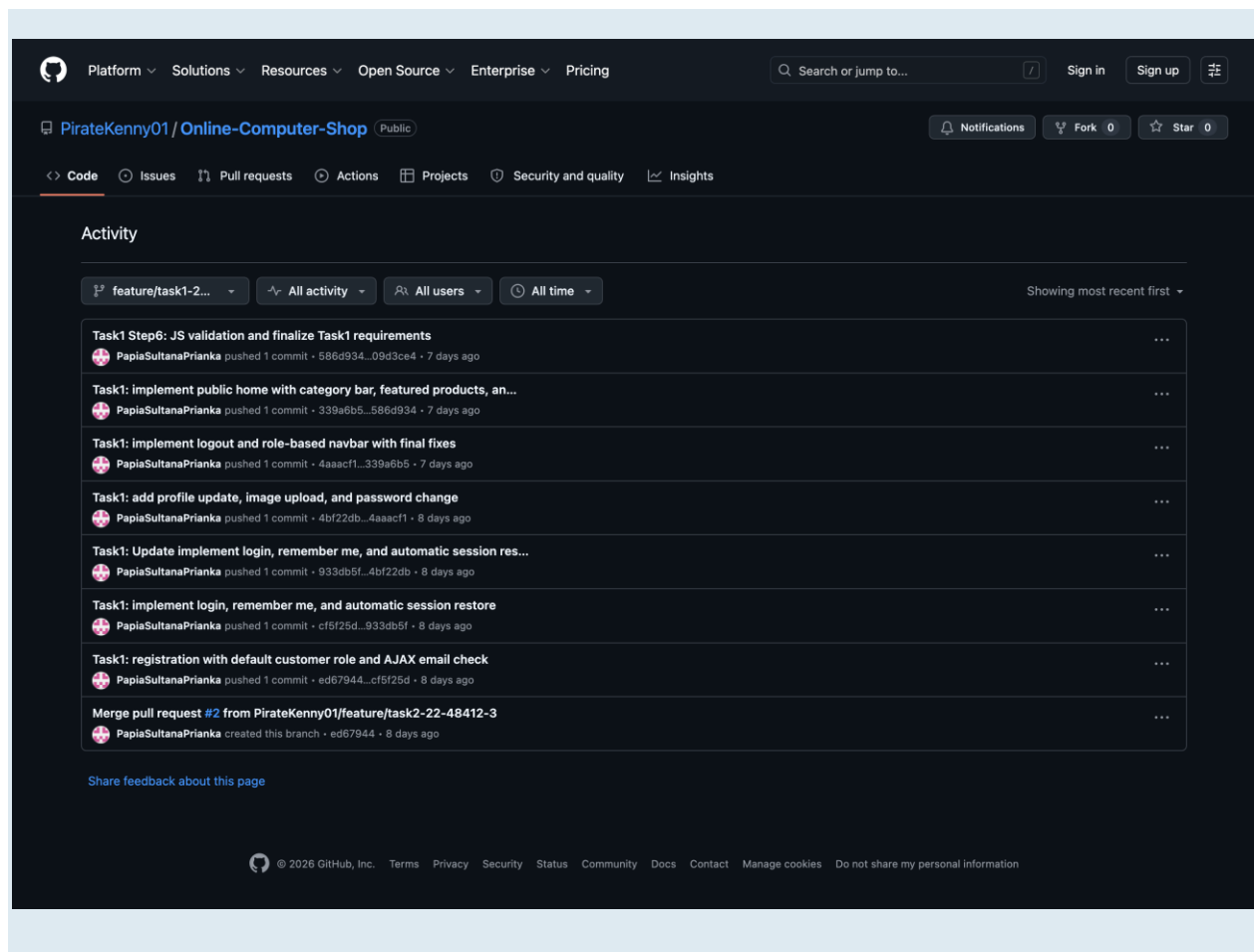
2.1 GitHub Repository Link

Project Repository URL: <https://github.com/PirateKenny01/Online-Computer-Shop.git>

My Repository URL: <https://github.com/PapiaSultanaPrianka/learning-web-technologies-spring2025-2026-sec-d.git>

My Branch: feature/task1-22-48356-3

2.2 Git Contribution Screenshot



The screenshot shows a GitHub repository page for the project "learning-web-technologies-spring2025-2026-sec-d" by user "PapiaSultanaPrianka". The repository is public and has 27 commits. The file list includes folders for "CRUD_OPERATION", "CRUD_OPERATION_Full_Code", "Final lab 1", "LAB-7.2[HTML-Form][JS]", "Lab1", "Lab2", "Online-Computer-Shop", "Product", and files for ".gitignore", "Labexam.html", "Labexam.js", and "README.md". The README file is selected, showing the repository name. The right sidebar contains sections for "About", "Releases", "Packages", "Contributors", "Languages", and "Suggested workflows". The "Languages" section shows a bar chart with PHP (61.1%), JavaScript (8.3%), CSS (0.3%), HTML (28.9%), and Hack (1.4%). The "Suggested workflows" section lists "Symfony", "SLSA Generic generator", and "Datadog Synthetics".

Repository: learning-web-technologies-spring2025-2026-sec-d (Public)

Files:

File/Folder	Commit Message	Time Ago
CRUD_OPERATION	CREATE OPERATION	3 weeks ago
CRUD_OPERATION_Full_Code	CRUD Full LAB Task	3 weeks ago
Final lab 1	add	last month
LAB-7.2[HTML-Form][JS]	add	3 months ago
Lab1	ADD	3 months ago
Lab2	ADD	3 months ago
Online-Computer-Shop	Push Online-Computer-Shop project files	2 days ago
Product	add	2 months ago
.gitignore	Ignore DS_Store	2 days ago
Labexam.html	Add files via upload	2 months ago
Labexam.js	ADD	2 months ago
README.md	Initial commit	3 months ago

README

learning-web-technologies-spring2025-2026-sec-d

About

No description, website, or topics provided.

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

Contributors 1

Languages

Language	Percentage
PHP	61.1%
JavaScript	8.3%
CSS	0.3%
HTML	28.9%
Hack	1.4%

Suggested workflows

Based on your tech stack

- Symfony** [Configure](#)
Test a Symfony project.
- SLSA Generic generator** [Configure](#)
Generate SLSA3 provenance for your existing release workflows
- Datadog Synthetics** [Configure](#)
Run Datadog Synthetic tests within your GitHub Actions workflow

[More workflows](#) [Dismiss suggestions](#)

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Chapter 3: Features, Source Code & Screenshots

Feature: User Registration

Description:

- This feature allows a new visitor to create an account using name, email, and password.
- Its purpose is to populate the users table securely and route the user to login after successful registration.

Relevant Source Code:

- File: views/signup.php

```
<form method="post" action="../../controllers/signupCheck.php" onsubmit="return validateSignup(">
  Name: <input type="text" id="name" name="name" value="<?php if(isset($old['name'])) {echo
htmlspecialchars($old['name']);} ?>"/> <br>
  Email: <input type="text" id="email" name="email" onkeyup="checkEmailAvailability()"
value="<?php if(isset($old['email'])) {echo htmlspecialchars($old['email']);} ?>"/> <br>
  <p id="email_msg" style="color:red;"><?php if(isset($errors['email']) ||
isset($errors['email_exists'])) {echo isset($errors['email']) ? $errors['email'] : $errors['email_exists'];}
?></p>

  Password: <input type="password" id="password" name="password" value=""/> <br>
  Confirm Password: <input type="password" id="confirm_password" name="confirm_password"
value=""/> <br>
  <input type="submit" name="signup_submit" value="Signup"/>
</form>

<script src="../../public/js/main.js"></script>
```

- File: controllers/signupCheck.php

```
if ($name == "" || $email == "" || $password == "" || $confirm == "") {
    $errors['form'] = "null input found!";
}
if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
    $errors['email'] = "invalid email format!";
}
if (strlen($password) < 8) {
    $errors['password'] = "password must be at least 8 characters!";
}
if ($password != $confirm) {
    $errors['confirm_password'] = "password and confirm password mismatch!";
}
if (emailExists($email)) {
    $errors['email_exists'] = "email already exists!";
}
```

```
$user = [  
    'name' => $name,  
    'email' => $email,  
    'password_hash' => password_hash($password, PASSWORD_DEFAULT),  
    'role' => 'customer'  
];  
  
if (addUser($user)) {  
    $_SESSION['msg'] = "registration successful. please login.";  
    header('location: ../views/login.php');  
}
```

- File: api/checkEmail.php

```
$email = trim($_GET['email']);  
  
if ($email == "" || !filter_var($email, FILTER_VALIDATE_EMAIL)) {  
    echo json_encode(['status' => false, 'message' => 'invalid email']);  
    exit();  
}  
  
if (emailExists($email)) {  
    echo json_encode(['status' => true, 'available' => false, 'message' => 'email already exists']);  
} else {  
    echo json_encode(['status' => true, 'available' => true, 'message' => 'email available']);  
}
```

Code Explanation:

- Server-side validation checks null input, email format, password length, and confirm password match.
- Duplicate email is prevented through emailExists(\$email) before insert.
- Password is never stored in plain text; password_hash(..., PASSWORD_DEFAULT) is used.
- On success, user is redirected to login with a flash message in session.
- Client-side JS (validateSignup) and AJAX (checkEmailAvailability) provide early feedback before final server validation.

[Home](#) | [Login](#) | [Signup](#)

Signup

Name:

Email:

Password:

Confirm Password:

[Login](#)

[Home](#) | [Profile](#) | [Admin Create User](#) | [Logout](#)

Admin Dashboard - Create User

Name:

Email:

Password:

Confirm Password:

Role: 

[Back](#)

Feature: User Login + Remember Me

Description:

- This feature authenticates users and creates session identity (user_id, name, role).

- Its purpose is to support both normal login and persistent login across visits using a secure remember token.

Relevant Source Code:

- File: controllers/loginCheck.php

```
$user = getUserByEmail($email);

if ($user && password_verify($password, $user['password_hash'])) {
    $_SESSION['user_id'] = $user['id'];
    $_SESSION['name'] = $user['name'];
    $_SESSION['role'] = $user['role'];

    if ($remember == "yes") {
        $token = bin2hex(random_bytes(32));
        $token_hash = hash('sha256', $token);
        updateRememberToken($user['id'], $token_hash);
        setcookie('remember_token', $token, time() + 60 * 60 * 24 * 30, '/');
    } else {
        clearRememberToken($user['id']);
        setcookie('remember_token', "", time() - 10, '/');
    }

    header('location: ../views/home.php');
}
```

- File: views/login.php

```
if (!isset($_SESSION['user_id']) && isset($_COOKIE['remember_token'])) {
    $token_hash = hash('sha256', $_COOKIE['remember_token']);
    $user = getUserByRememberToken($token_hash);

    if ($user) {
        $_SESSION['user_id'] = $user['id'];
        $_SESSION['name'] = $user['name'];
        $_SESSION['role'] = $user['role'];
        header('location: home.php');
        exit();
    } else {
        setcookie('remember_token', "", time() - 10, '/');
    }
}
```

- File: models/userModel.php

```
function updateRememberToken($id, $token_hash)
{
    $sql = "update users set remember_token=? where id=?";
```

```
...
}  
  
function getUserByRememberToken($token_hash)  
{  
    $sql = "select id, name, role from users where remember_token=? limit 1";  
    ...  
}
```

Code Explanation:

- password_verify checks login password against hashed DB password.
- Session variables are set immediately after successful authentication.
- For “Remember Me,” a cryptographically strong token is generated (random_bytes), hashed, stored in DB, and raw token is stored in cookie for 30 days.
- On later visits, cookie token is hashed again and matched with DB token to auto-restore session.
- Invalid/expired token is cleaned from cookie to prevent repeated failed restoration.

[Home](#) | [Login](#) | [Signup](#)

Login

Email:
Password:
Remember Me: ☐
 [Signup](#)

Feature: Profile Management

Description:

- This feature provides a session-protected page where users can edit name/email, upload profile picture, and change password.
- Its purpose is to support user profile read/update operations securely (Task 1 CRUD scope).

Relevant Source Code:

- File: views/profile.php

```
if (!isset($_SESSION['user_id'])) {  
    if (isset($_COOKIE['remember_token'])) {  
        $token_hash = hash('sha256', $_COOKIE['remember_token']);  
        $rememberUser = getUserByRememberToken($token_hash);  
        ...  
    } else {  
        header('location: login.php');  
        exit();  
    }  
}
```

```
<p style="color:green;"><?php if (isset($_SESSION['profile_success'])) { echo  
$_SESSION['profile_success']; unset($_SESSION['profile_success']); } ?></p>
```

```
<form method="post" action="../controllers/profileUpdateCheck.php" enctype="multipart/form-data"  
onsubmit="return validateProfileUpdate()">  
...  
</form>
```

```
<form method="post" action="../controllers/passwordChangeCheck.php" onsubmit="return  
validatePasswordChange()">  
...  
</form>
```

- File: controllers/profileUpdateCheck.php

```
if($email != "" && !filter_var($email, FILTER_VALIDATE_EMAIL)){  
    $errors['email'] = "invalid email format!";  
}  
if(emailExistsForOther($email, $id)){  
    $errors['email_exists'] = "email already used by another user!";  
}  
  
if(isset($_FILES['profile_picture']) && $_FILES['profile_picture']['name'] != ""){  
    ...  
    if($size > 2*1024*1024){ ... }  
    $mime = mime_content_type($tmp);  
    $allowed = ['image/jpeg' => 'jpg', 'image/jpg' => 'jpg', 'image/png' => 'png'];  
    ...  
    $uploadDirFs = __DIR__.'../public/uploads/';  
    ...  
}
```

```
if(move_uploaded_file($tmp, $uploadPathFs)){
    $profile_picture = "public/uploads/".$newName;
}
}

if(updateUserProfile($user)){
    $_SESSION['name'] = $name;
    $_SESSION['profile_success'] = "profile updated successfully.";
}
```

- File: controllers/passwordChangeCheck.php

```
if(strlen($new_password) < 8){
    $errors['new_password'] = "new password must be at least 8 characters!";
}
if($new_password != $confirm_password){
    $errors['confirm_password'] = "new and confirm password mismatch!";
}

$user = getUserById($id);
if($user && !password_verify($current_password, $user['password_hash'])){
    $errors['current_password'] = "current password not matched!";
}

$new_hash = password_hash($new_password, PASSWORD_DEFAULT);
updateUserPassword($id, $new_hash);
```

Code Explanation:

- Access is session-gated; unauthenticated users are redirected to login.
- Profile update validates null input, valid email format, and unique email excluding current user.
- Image upload enforces file size limit (2MB), MIME allowlist (JPG/PNG), and stores files in public/uploads/.
- Password change requires current password verification before hashing/storing new password.
- Success messages are returned via session flash (profile_success).
- Client-side functions validateProfileUpdate() and validatePasswordChange() in public/js/main.js enforce early checks.

[Home](#) | [Profile](#) | [Logout](#)

Profile

Update Profile



Name:
 Email:
 Profile Picture: no file selected

Change Password

Current Password:
 New Password:
 Confirm Password:

Feature: Navbar & Logout

Description:

- This feature displays role-specific navigation links and supports full user sign-out.
- Its purpose is to provide role-aware access points and ensure session/token cleanup during logout.

Relevant Source Code:

- File: views/partials/navbar.php

```
$role = isset($_SESSION['role']) ? $_SESSION['role'] : 'guest';
```

```
<?php if ($role == 'admin') { ?>
  <a href="home.php">Home</a> |
  <a href="profile.php">Profile</a> |
  <a href="admin_create_user.php">Admin Create User</a> |
  <a href=" ../controllers/logout.php">Logout</a>
<?php } else if ($role == 'customer') { ?>
  <a href="home.php">Home</a> |
  <a href="profile.php">Profile</a> |
  <a href=" ../controllers/logout.php">Logout</a>
<?php } else { ?>
  <a href="home.php">Home</a> |
  <a href="login.php">Login</a> |
```

```
<a href="signup.php">Signup</a>
<?php } ?>
```

- File: controllers/logout.php

```
if (isset($_SESSION['user_id'])) {
    clearRememberToken($_SESSION['user_id']);
} else if (isset($_COOKIE['remember_token'])) {
    $token_hash = hash('sha256', $_COOKIE['remember_token']);
    $u = getUserByRememberToken($token_hash);
    if ($u) { clearRememberToken($u['id']); }
}

$_SESSION = [];
setcookie('remember_token', "", time() - 10, '/');
setcookie(session_name(), "", time() - 10, '/');
session_destroy();

header('location: ../views/login.php');
```

Code Explanation:

- Navbar resolves current role from session and branches links for admin, customer, and guest.
- Logout removes persistent remember token from database first.
- All runtime auth state is cleared: session array, remember cookie, and PHP session cookie.
- Final redirect sends the user to login page, preventing authenticated page reuse.

[Home](#) | [Profile](#) | [Admin Create User](#) | [Logout](#)

Online Computer Shop

Category Bar

[Graphics Card](#) | [Memory](#) | [Monitor](#) | [Processor](#) | [Storage](#) |

Featured Components

Name	Manufacturer Review	Price
Intel Core i5 12400F	Great performance for gaming and daily productivity.	21,999.00
AMD Ryzen 5 5600	Excellent value CPU with strong multi-core result.	18,999.00
NVIDIA RTX 4060	Efficient GPU for 1080p ultra gaming.	42,999.00
Corsair Vengeance 16GB DDR4	Stable and reliable RAM for mainstream builds.	6,499.00
Samsung 980 1TB NVMe SSD	Fast boot and load time, solid reliability.	9,999.00
LG 24MP400 24-inch Monitor	Good color and viewing angle for budget users.	15,999.00

[Home](#) | [Profile](#) | [Logout](#)

Online Computer Shop

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[Home](#) | [Login](#) | [Signup](#)

Online Computer Shop

Category Bar

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Featured Components

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Feature: Home Page

Description:

- This feature is the public landing page and central display page for Task 1.
- Its purpose is to aggregate dynamic category and featured product data for all visitors.

Relevant Source Code:

- File: index.php

```
header('location: views/home.php');  
exit();
```

- File: views/home.php

```
session_start();  
require_once('../models/userModel.php');  
require_once('../models/categoryModel.php');  
require_once('../models/productModel.php');  
  
if (!isset($_SESSION['user_id']) && isset($_COOKIE['remember_token'])) {  
    $token_hash = hash('sha256', $_COOKIE['remember_token']);  
    $user = getUserByRememberToken($token_hash);
```

```
if ($user) {
    $_SESSION['user_id'] = $user['id'];
    $_SESSION['name'] = $user['name'];
    $_SESSION['role'] = $user['role'];
} else {
    setcookie('remember_token', "", time() - 10, '/');
}
}

$categories = getTopCategories();
$featuredProducts = getFeaturedProducts();
```

Code Explanation:

- Root URL forwards to views/home.php, making Home the default entry point.
- Home loads required models and fetches dynamic data for categories and products.
- If a valid remember cookie exists, session is automatically re-established for personalized navbar behavior.
- This page is intentionally accessible to guests as well as logged-in users.

Feature: Category Bar

Description:

- This feature shows all top-level categories in navigation format and supports category-specific browsing.
- Its purpose is to provide structured product discovery from the database taxonomy.

Relevant Source Code:

- File: models/categoryModel.php

```
function getTopCategories()
{
    $con = getConnection();
    $sql = "select id, name from categories where parent_id is null order by name asc";
    $result = mysqli_query($con, $sql);

    $categories = [];
    while ($row = mysqli_fetch_assoc($result)) {
        $categories[] = $row;
    }
    return $categories;
}
```

```
}
```

- File: views/home.php

```
<h3>Category Bar</h3>
<div>
  <?php if (count($categories) > 0) { ?>
    <?php foreach ($categories as $cat) { ?>
      <a href="category.php?id=<?php echo $cat['id']; ?>"><?php echo htmlspecialchars($cat['name']);
?></a> |
    <?php } ?>
  <?php } ?>
</div>
```

- File: views/category.php

```
$categoryId = 0;
if (isset($_GET['id'])) {
    $categoryId = (int)$_GET['id'];
}

if ($categoryId > 0) {
    $selectedCategory = getCategoryById($categoryId);
    if ($selectedCategory) {
        $products = getProductsByTopCategory($categoryId);
    }
}
```

Code Explanation:

- Category list is DB-driven and filtered to only top-level categories (parent_id is null).
- Each category link passes its ID to category.php through query string.
- category.php sanitizes with integer casting and fetches selected category plus products under that top category/subcategories.
- This separates navigation concern (view) from query logic (model), preserving MVC structure.

[Home](#) | [Login](#) | [Signup](#)

Online Computer Shop

Category Bar

[Graphics Card](#) | [Memory](#) | [Monitor](#) | [Processor](#) | [Storage](#) |

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LG 24MP400 24-inch Monitor	Good color and viewing angle for budget users.	15,999.00

Feature: Featured Products

Description:

- This feature displays a compact list of highlighted products on Home.
- Its purpose is to show recent/important items with quick decision data: name, manufacturer review snippet, and price.

Relevant Source Code:

- File: models/productModel.php

```
function getFeaturedProducts()
{
    $con = getConnection();
    $sql = "select id, name, manufacturer_review, price, category_id from products order by created_at
desc limit 6";
    $result = mysqli_query($con, $sql);

    $products = [];
    while ($row = mysqli_fetch_assoc($result)) {
        $products[] = $row;
    }
}
```

```
return $products;
}
```

- File: views/home.php

```
<h3>Featured Components</h3>
<?php if (count($featuredProducts) > 0) { ?>
  <table border="1" cellpadding="8" cellspacing="0">
    <tr>
      <th>Name</th>
      <th>Manufacturer Review</th>
      <th>Price</th>
    </tr>
    <?php foreach ($featuredProducts as $p) { ?>
      <tr>
        <td><?php echo htmlspecialchars($p['name']); ?></td>
        <td>
          <?php
            $review = trim((string)$p['manufacturer_review']);
            if ($review == "") { echo "No review available"; }
            else {
              $short = substr($review, 0, 100);
              if (strlen($review) > 100) { $short .= "..."; }
              echo htmlspecialchars($short);
            }
          ?>
        </td>
        <td><?php echo number_format((float)$p['price'], 2); ?></td>
      </tr>
    <?php } ?>
  </table>
<?php } ?>
```

Code Explanation:

- Featured product data is fetched from products ordered by latest created_at, limited to 6 rows.
- The view renders required Task 1 columns: product name, short manufacturer review, and formatted price.
- Review text is safely handled with trimming, empty fallback message, and 100-character truncation.
- Output escaping (htmlspecialchars) and numeric formatting (number_format) improve display safety and consistency.

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Online Computer Shop

Category Bar

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Featured Components

Name	Manufacturer Review	Price
Intel Core i5 12400F	Great performance for gaming and daily productivity.	21,999.00
AMD Ryzen 5 5600	Excellent value CPU with strong multi-core result.	18,999.00
NVIDIA RTX 4060	Efficient GPU for 1080p ultra gaming.	42,999.00
Corsair Vengeance 16GB DDR4	Stable and reliable RAM for mainstream builds.	6,499.00
Samsung 980 1TB NVMe SSD	Fast boot and load time, solid reliability.	9,999.00
LG 24MP400 24-inch Monitor	Good color and viewing angle for budget users.	15,999.00

Chapter 4: Conclusion & Future Work

5.1 Summary

This project successfully implemented an Online Computer Shop system. My contribution focused on authentication, homepage design, and user profile management. All features were developed following MVC architecture and secure coding practices.

5.2 Challenges Faced

- Implementing secure password hashing and verification
- Managing session and “Remember Me” functionality
- Handling file uploads for profile pictures
- Maintaining MVC structure properly

5.3 Future Improvements

- Add email verification during registration
- Improve UI with modern frameworks (React/Bootstrap)
- Add two-factor authentication
- Optimize performance and database queries

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